Total No. of Questions : 8]

SEAT No. :

PC2821

[Total No. of Pages : 4

[6352]-45

S.E. (Artificial Intelligence & Data Science)

STATISTICS

(2019 Pattern) (Semester - IV) (217528)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Figures to the right indicate full marks.*
3) *Neat diagrams must be drawn wherever necessary.*
4) *Make suitable assumption whenever necessary.*

Q1) a) Explain the following terms : **[8]**

i) Skewness & Kurtosis

ii) Regression

b) A collar manufacturer is considering the production of a new style collar to attract young men. The following statistics of neck circumference are available based on the measurement of a typical group of students. Calculate mean and standard deviation. **[10]**

	12.5	13	13.5	14	14.5	15	15.5	16
Frequency	4	19	30	63	66	29	18	1

OR

Q2) a) The following marks have been obtained by a class of students in 2 papers of mathematics. **[9]**

Paper I	45	55	56	58	60	65	68	70	75	80	85
Paper II	56	50	48	60	62	64	65	70	74	82	90

Calculate the coefficient of correlation for the above data.

b) Determine the equations of regression lines for the following data. Also find the values of **[9]**

i) y for $x = 4.5$

ii) x when $y = 13$

X	2	3	5	7	9	10	12	15
Y	2	5	8	10	12	14	15	16

P.T.O.

Q3) a) 10% of bolts produced by a machine are defective. Determine the probability that out of 10 bolts chosen at a random. **[5]**

i) 2 will be defective

ii) at most 2 will be defective

b) A dice is thrown 10 times. If getting an odd number is a success, What is the probability of **[6]**

i) 8 success

ii) at least 6 success

c) For a normal distribution When mean = 2, standard deviation $\sigma = 4$, find the probabilities of the following intervals. **[6]**

i) $4.43 \le x \le 7.29$

ii) $-0.43 \le x \le 5.39$

[Given : A (z = 0.61) = 0.2291, A(z = 1.32) = 0.4066, A (z = 0.85) = 0.3023]

OR

Q4) a) A random variable X has the following probability distribution : **[5]**

i) Find k

ii) Evaluate P (X < 6)

iii) P(X > = 6)

iv) P(0 < X < 5)

X	0	1	2	3	4	5	6	7
P(X)	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

b) Fit a Poisson distribution to the following data and calculate theoretical frequencies. **[6]**

x	0	1	2	3	4	Total
f	109	65	22	3	1	200

c) The lifetime of an article has a normal distribution with mean 400 hours and standard deviation 50 hours. Find the expected number of articles out of 2000 whose lifetime lies between 335 hours to 465 hours. **[6]**

(Given : A (z = 1.3) = 0.4032)

Q5) a) In a Batch of 500 articles, produced by a machine, 16 articles are found defective. After overhauling the machine, it is found that 3 articles are defective in a batch of 100. Has the machine improved? **[6]**

(Given $Z_\alpha = 1.96$)

b) The Table below gives the number of customers visit the certain company on various days of week

Days	Sun	Mon	Tue	Wed	Thurs	Fri	Sat
Number of Customers	6	4	9	7	8	10	12

Test at 5% of level of significance whether customer visits are uniformly distributed over the days. **[6]**

[Given $\chi^2_{6,0.05} = 15.592$]

c) Two random samples gave the following results. Test whether the samples came from the same normal population at 5% level of significance. **[6]**

Sample	Size	Sample mean	Sum of squares of deviations from them mean
1	10	15	90
2	12	14	108

Given : $F(0.05, (9, 11)) = 2.9$

OR

Q6) a) A survey of 320 families with 5 children each revealed the following distribution : **[9]**

No. of boys	0	1	2	3	4	5
No. of family	12	40	88	110	56	14

Use chi-square test to test the hypothesis that data follows a binomial distribution (chi-square = 11.07 at 5% level of significance)

b) A random sample of 10 boys had the following I.Q.'s : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100. Given : $t(0.05,9) = 2.262$. **[9]**

Q7) a) Let P be the probability that a coin will fall head in a single toss in order to test $H_0 : P = \frac{1}{2}$ against $P = \frac{3}{4}$. The coin is tossed 5 times and H_0 is rejected if more than 3 heads are obtained. Find the probability of type I error and power of the test. **[8]**

b) What do you mean by Non Parametric Test? State its advantages & disadvantages. **[9]**

OR

Q8) a) Write short note on : **[8]**

i) Population and sample

ii) Type I and Type II error

iii) Critical Region

iv) Power of test

b) State and prove Neyman-Pearson lemma for testing a simple hypothesis against a simple alternative hypothesis. **[9]**